FOUR-DIGIT ACCURACY OF THE MONOTONE-DELTA CONSTANT

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The upper bound is sharp "at least up to the first four significant digits."

The paragraph after Theorem 1, Section 1.2, p. 2, of Andrés Chirre, Dimitar K. Dimitrov, Emily Quesada-Herrera and Mateus Sousa, *An extremal problem and inequalities for entire functions of exponential type*, doi:10.1090/proc/16764, Proc. Amer. Math. Soc. 152 (2024). The constant is $A_1 = \inf_f \|f\|_1/f(0)$ over nonnegative radially decreasing real entire functions of exponential type at most 2π.

Answered: yes. Evidence: Sketch.

$1.2771345 < A_1 < 1.277135042105$, which fixes the requested four significant digits.

Proof. The source reduces the extremal problem to the reciprocal norm of a positive Gram operator Q on ℓ^2. Let P_d project onto the first d coordinates and write

$$Q = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}, \qquad \lambda_d = \|A\|.$$

The published estimates on the matrix entries give $\|B\| < 3/(2d)$ and $\|C\| < 1/d$ when $d = 3010$, while the source's certified finite-matrix calculation gives $\lambda_d < \ell = 783002554181/10^{12}$. For a unit vector (x, y) in the two blocks,

$$\langle Q(x,y),(x,y)\rangle \le \ell\|x\|^2 + 2b\|x\|\|y\| + c\|y\|^2, \qquad b = \frac{3}{2d}, \quad c = \frac{1}{d}.$$

The largest eigenvalue of this 2×2 form is at most $\ell + b^2/(\ell - c)$, and exact rational comparison at $d = 3010$ yields $\|Q\| < 10^7/12771345$. Since $A_1 = 1/\|Q\|$, the lower bound follows. The published certified trial function gives the upper bound 1.277135042105. This proves the stated enclosure, not the stronger nine-digit conjecture elsewhere in the paper. □

AI assistance was used in adapting and checking the argument and the exposition.

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